

Ray Tsai

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EDUCATION

University of Oxford

Oxford, United Kingdom

M.Sc. in Mathematical Sciences

Sep 2025 – Jun 2026

- **Dissertation:** *Induced Subgraphs in Ramsey Graphs*, advised by Prof. Alex Scott
- **Relevant coursework:** Analytic Number Theory, Theories of Deep Learning, Probability on Graphs and Lattices, Combinatorics, Probabilistic Combinatorics, Networks, Continuous Optimization

University of California San Diego

La Jolla, CA

B.Sc. in Mathematics-Computer Science (Honors with Highest Distinction), Math GPA: 3.98/4.0

Sep 2021 – Jun 2025

- **Honors Thesis:** *Double Turán Problems*, advised by Prof. Jacques Verstraete
- **Selected coursework:** Extremal Combinatorics and Graph Theory (Honors), Algebraic Combinatorics (Graduate), Discrete Geometry (Graduate), Statistical Learning (Graduate), Complex Analysis (Graduate)

AWARDS

Physical Sciences Dean's Undergraduate Award for Excellence

UC San Diego, 2025

- “For demonstrated academic excellence and promise as researchers in UC San Diego’s three areas of study: Chemistry and Biochemistry, Mathematics, and Physics.” (\$1000)

RESEARCH EXPERIENCE

UC San Diego, Department of Mathematics

La Jolla, CA

Undergraduate Research Assistant

Mar 2023 – Jun 2025

- Studied various problems and related literature in extremal graph theory under Prof. Jacques Verstraete.
- Completed honors thesis on *Double Turán Problems*, earning the program’s highest distinction as one of approximately 12 students annually to complete the Math Honors Program.

UC San Diego, Department of CSE

La Jolla, CA

Undergraduate Research Assistant

Apr 2024 – Jun 2025

- Worked through graduate-level complexity theory textbook directly under Prof. Russell Impagliazzo.
- Joined Professor Impagliazzo’s research group on smooth boosting; studied the connections between algorithmic fairness, computational complexity, and learning theory.

TEACHING EXPERIENCE

Instructional Assistant

La Jolla, CA

UC San Diego, Department of CSE

Jan 2025 – Mar 2025

- Supported Design & Analysis of Algorithms (CSE 101) with 200+ students; conducted weekly office hours and graded assignments on algorithmic concepts including dynamic programming, graph algorithms, and NP-completeness.

WORK EXPERIENCE

MSI

New Taipei City, Taiwan

Software Engineer

Jul 2023 – Aug 2023, Full-time

- Developed Windows application for local AI image generation with 5-member team; fine-tuned Stable Diffusion model to generate photorealistic MSI laptop images for marketing materials.
- Engineered automated installation pipeline for optimization backends, improving image generation throughput by 80%.

Kook

San Diego, CA

Co-founder & Lead Software Engineer

Dec 2021 – Dec 2022, Start-up

- Co-founded hardware-software startup for automated surfboard rental stations; secured UCSD Basement incubation and led 5-member engineering team through product development lifecycle.
- Architected full-stack React Native application with NFC-based authentication for 100+ beta users, reducing rental initiation time by 75%.

TECHNICAL SKILLS

Languages: Python, C/C++, Java, TypeScript, LaTeX, SQL **Frameworks:** PyTorch, React, React Native, Flask